

SPCE 5065

MIDTERM

You are allowed any course notes, the textbooks by Tribble and/or Pisacane, any material on the Canvas page, and a calculator.

I completed this assignment using only authorized references

Signed: _____

Printed: _____

Problem 1: _____/10

Problem 2: _____/10

Problem 3: _____/15

Problem 4: _____/15

Problem 5: _____/10

Problem 6: _____/20

Problem 7: _____/20

Total: _____

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1. (10 pts.) True/False

- a. FALSE Drag reduces an orbit's total and kinetic energy.
- b. FALSE Newton's second law states that $\sum \vec{F} = m\vec{a}$, which is directly applicable to rockets.
- c. TRUE The coldest layer in the atmosphere is the mesosphere.
- d. TRUE When the frequency of galactic cosmic ray events is low, the frequency of extreme solar events is high.
- e. FALSE Drag acting on a satellite in LEO reduces its altitude linearly.
- f. FALSE Atomic oxygen is the main constituent in the heterosphere during solar min.
- g. TRUE Astronauts tend to eat less in free-fall.
- h. TRUE Earth's magnetic field axis is constantly moving.
- i. FALSE Our current sun cycle (25) is closely matched predications.
- j. FALSE GEO satellites do not need protection from the space environment because there is no atmosphere there.

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2. (10 pts.)

I. Which of the effects are *not* techniques to mitigate the neutral environment effects
(select all that apply)

- a. Choose materials resistant to atomic oxygen erosion
- b. Shield sensitive surfaces
- ☒ c. Bias the spacecraft with a positive voltage
- d. Use coatings on sensitive surfaces
- ☒ e. Choose orbits where the most objects are located

II. A mission designer must choose *one* improvement for a CubeSat expected to operate for five years at 500 km altitude. Which option would provide the greatest increase in mission lifetime?

- ☒ a. Reduce the spacecraft's frontal area by 40%.
- b. Replace the flight computer with a radiation-hardened processor.
- c. Increase solar array area by 30%.
- d. Add an additional communications antenna.
- e. Increase battery capacity by 50%.

III. Solar ultraviolet radiation alters which of the following, which results in temperature changes on the spacecraft (choose the *best* answer)?

- a. strength to flexibility ratio
- b. reflectivity to transmissivity ratio
- ☒ c. absorptivity to emissivity ratio
- d. stress to strain ratio
- e. decomposition to desorption ratio

IV. A Mars transit vehicle is forced to reduce mass by eliminating *one* of the following:

- psychological screening
- simulator training
- exercise equipment
- habitable volume
- medical diagnostic equipment

Which reduction is most likely to produce the greatest increase in overall mission risk?

- a. Psychological screening
- b. Simulator training
- ☒ c. Exercise equipment
- d. Habitable volume
- d. Medical diagnostic equipment

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V. What is the primary *cause* of solar flares?

- a. Nuclear fusion in the Sun's core
- ☒ b. Magnetic field reconnection
- c. Cosmic ray interactions
- d. Sunspot cooling
- e. Coronal mass ejections

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3. (15 pts) A four-person Mars mission has limited mass available. The project manager proposes eliminating one of the following to reduce launch mass:

- astronaut psychological screening,
- long-duration team training,
- simulator training.

Which would you recommend keeping and why?

Discuss at least **three** ways your choice contributes to mission success, using examples from astronaut selection, stress and coping, group dynamics, or previous space missions.

I would keep long duration team training. For a 4 person crew locked in an isolated, confined environment for ~2.5 years with no evacuation and 20+ minute communication delays, the crew succeeds or fails as a team, and team behavior is the biggest driver that can still be purchased at this point. The three ways it earns its mass:

1. Group dynamics/cohesion: Isolated confined environment (ICE) analogs (Mars500, Antarctic winter-over, ISS expeditions) consistently show that team friction, not hardware, is what degrades long missions. Long-duration training builds shared mental models, communication habits, and conflict-resolution reflexes that only form with time together [3].

2. Stress and coping. A team that has trained together has pre-negotiated roles and coping strategies, which blunts the "third-quarter" morale dip and keeps decision-making intact when a real emergency hits. Shuttle-Mir showed the opposite: crews that had not trained to a common standard hit language and expectation seams that cost performance [3].

3. Recovery from the loss of the other two. Good team training partially covers for thinner screening (a cohesive team self-monitors and manages a struggling member) and for less simulator time (a coordinated crew can work procedures collaboratively en route, and Mars-transit downtime is long). The reverse is not true: a perfectly screened but un-gelled crew still has to learn to operate together somewhere, and deep space is the worst classroom.

The tradeoff I am accepting: psychological screening is the close runner-up (you cannot fully train away a fundamentally incompatible crewmember), and simulator training is the most deferrable because procedures can be practiced during the long transit. If I could only protect one lever on team performance, it is the training that actually forges the team.

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4. (15 pts) Your spacecraft has just entered safe mode. Telemetry shows:

- the spacecraft is healthy
- battery voltage is nominal
- multiple computer resets occurred over the South Atlantic
- NOAA issued a G3 geomagnetic storm warning the previous day

Explain:

- a. the most likely cause
- b. what additional data you would request
- c. the immediate operational actions you would take
- d. the long-term design improvements (for future mission) you would recommend

a. Most likely cause: radiation-induced single-event effects driven by trapped protons in the South Atlantic Anomaly and amplified by the G3 storm. A healthy bus and nominal battery rule out power and thermal, so this is not a hardware or eclipse problem. Resets that cluster over the South Atlantic are the textbook SAA signature (the inner belt dips low there, so proton flux spikes), and a G3 geomagnetic storm the day before pumps up energetic particle populations, raising the upset rate [1]. Bottom line: energetic particles are flipping bits and latching logic, tripping the watchdog into safe mode.

b. Additional data I would request:

- Onboard error logs: which unit reset, EDAC single/multi-bit error counts, memory-scrub history.
- Reset timestamps cross-correlated with ground track to confirm the SAA overlap.
- Space-weather data: GOES proton/electron flux, Kp/Dst for the G3 event, timeline vs the resets.
- Dosimeter / particle-detector telemetry if the bus carries one, to see the local flux during each event.

c. Immediate operational actions:

- Stay in safe mode until the storm subsides and flux returns to baseline.
- Command a memory scrub and reload from a known-good image; clear and re-arm error counters.
- Inhibit critical activities (maneuvers, sensitive science) during SAA passes for now.
- If any unit is latched (SEL), power-cycle it to clear the latch-up before it does thermal damage, then verify health.

d. Long-term design improvements:

- Rad-hardened or rad-tolerant processor and memory, with EDAC/error-correcting memory and periodic scrubbing.
- Watchdog plus autonomous recovery so a hung computer resets and recovers itself instead of sitting in safe mode (the Galaxy 15 lesson: a latched, un-recovered bus turns a glitch into a mission-length saga) [1].
- Latch-up protection (current-limiting / power-cycle circuits) on susceptible parts.
- Targeted shielding of the avionics box and SAA-aware flight rules (safe sensitive ops over the anomaly), plus redundancy/voting on critical logic.

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5. (10 pts.) Initially the Starlink communication satellites were intended to be in LEO at an altitude of approximately 1100 km. Space X later decided to place the satellites at a lower (550 km) altitude due to concerns about the space environment. What might three of these concerns be?

Every one of these gets worse with altitude, which is why 550 km won:

- Radiation dose and SEUs. At 1100 km you are climbing into the bottom of the inner Van Allen belt, so trapped-proton and electron flux, total ionizing dose, and single-event rates all jump. That shortens electronics life and drives up shielding mass. At 550 km you sit well below the belt [1].
- Debris collision risk with no self-cleaning. At 1100 km atmospheric drag is negligible, so debris and dead satellites persist for centuries, and the collision/Kessler risk for a mega-constellation is severe. At 550 km the atmosphere naturally sweeps the band, so the environment self-cleans [1].
- End-of-life disposal. A failed satellite at 1100 km stays up for hundreds of years; the same failure at 550 km reenters on its own within roughly five years even with no propulsion (see Problem 6b), which is what responsible disposal and the debris-mitigation guidelines demand [1].

The price of going low is more drag and more atomic oxygen (Problem 6 is exactly that bill), but for a huge constellation the radiation and debris arguments dominate.

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6. (20 pts.)

a. Given the same Starlink satellite (at an altitude to 550 km), estimate the maximum erosion depth of a Mylar cover oriented on the RAM side during high solar activity over its five year mission life. If the thickness of the mylar cover is 300 μm , will this be a problem? Clearly state your assumptions. Assume an atomic oxygen density of 1×10^8 atoms/cm³.

$$\text{depth} = A_c F, \quad F = n_0 V t$$

$$\text{at } 550 \text{ km } r = 6928 \text{ km}, V = 7585 \text{ km/s} = 7.585 \times 10^5 \text{ cm/s}$$

$$\text{over } T = 5 \text{ yr} = 1.578 \times 10^8 \text{ s}$$

$$F = (1 \times 10^8) (7.585 \times 10^5) (1.578 \times 10^8) = 1.197 \times 10^{22} \text{ atoms/cm}^2$$

$$\text{depth} = (3.4 \times 10^{-24}) (1.197 \times 10^{22}) = 4.07 \times 10^{-2} \text{ cm}$$

erosion depth $\approx 407 \mu\text{m} > 300 \mu\text{m}$ cover
 $\therefore$ yes it is a problem.

b. Estimate the decay in altitude the satellite can expect over the five year mission life if no station-keeping is performed. Assume a ballistic coefficient of 103 kg/m². Use an average altitude of 350 km for your density calculations and assume an average R of 6728 km for this problem.

$$\frac{dR}{dt} = -\frac{\rho}{BC} \sqrt{\mu R}$$

Where: $\rho = 1.02 \times 10^{-12} \text{ kg/m}^3$

$$\text{b.) } \frac{dR}{dt} = -\frac{\rho}{BC} \sqrt{\mu R}$$

$$= -\frac{1.02 \times 10^{-12}}{103} \sqrt{3.986 \times 10^{14} (6.728 \times 10^6)}$$

$$= -2.91 \times 10^{-3} \text{ m/s}$$

$$\Delta R = (-2.91 \times 10^{-3}) (1.578 \times 10^8) = -4.59 \times 10^5 \text{ m}$$

c. If the thickness of the mylar cover is 50 μm and the deorbit altitude is 150 km, do you expect drag or mylar erosion will be the biggest concern for this mission?

c.) erosion 50/yr = .61 yr << drag to deorbit (4.4 yr)

atomic O₂ erosion is bigger concern.

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7. (20 pts.) You are the lead systems engineer for a 12U CubeSat (a 1U CubeSat is 10x10x10 cm) that must remain operational for five years in a 500 km circular orbit.

You may improve only **one** of the following:

- reduce frontal area by 40%,
- increase mass by 50%,
- reduce drag coefficient from 2.2 to 1.5.

Recommend one option and justify your decision using concepts from the neutral environment, ballistic coefficient, solar cycle variability, and orbital perturbations. Discuss any assumptions and tradeoffs.

Recommendation: reduce the frontal area by 40%. At 500 km the mission is drag-limited, and lifetime scales with the ballistic coefficient $BC=m/(C_d A)$, so the winning move is whichever option raises BC the most.

Ballistic coefficient (the deciding math). Each option multiplies BC by: area cut $\rightarrow \times 1/0.6=1.67$; mass up $\rightarrow \times 1.5$; drag-coefficient cut $\rightarrow \times 2.2/1.5=1.47$. Reducing frontal area gives the largest BC gain, so the longest life per the drag equation [1], [2].

Neutral environment. Drag is the dominant force at 500 km and the whole reason the orbit decays; shrinking the ram cross-section directly cuts the force ($F_{\text{drag}} \propto A$). It also cuts atomic-oxygen fluence on the ram face as a bonus, easing erosion [1].

Solar-cycle variability. A 5-year mission rides through a big chunk of the solar cycle, and density at 400-700 km swings by 10-30x from solar min to max [2]. The extra BC margin is exactly what buys survival through a solar-max density spike, when decay is worst.

Orbital perturbations. At 500 km drag is the perturbation that ends the mission; J_2 and third-body effects reshape the orbit but do not decay it. Spending the one improvement on the perturbation that actually kills you is the right call.

Why not the others: cutting C_d from 2.2 to 1.5 is the least achievable, real CubeSat drag coefficients sit around 2.2 to 4 in free-molecular flow (diffuse re-emission), so 1.5 is optimistic bordering on unphysical [2]. Increasing mass 50% is the simplest and most certain ($\times 1.5$, just add ballast), and if attitude control is shaky it is the safer pick, but it is a smaller BC gain and costs launch mass.

Assumptions and tradeoffs: I assume the 40% area reduction is realized by flying a minimum-cross-section attitude (or a slimmer deployed geometry) and holding it with the ADCS. The real cost is power and control: a smaller ram face can mean less sun-facing array area, and if attitude control drops out the satellite tumbles and the area (and drag) average back up, erasing

Sources Cited

[1] Tribble, A. C., The Space Environment: Implications for Spacecraft Design, rev. ed., Princeton Univ. Press, Princeton, NJ, 2003 (drag energetics, atmospheric structure, atomic oxygen, radiation/SAA, charging, thermal-control degradation).

[2] George, L., "SPCE 5065 Lesson Notes and Slides, Lessons 1 to 4 (The Space Environment; Neutral and Plasma Environments)," course material (Canvas), University of Colorado Colorado Springs, 2026 (atmospheric structure and the coldest-layer and drag-energy quizzes, ballistic- and drag-coefficient ranges, atomic-oxygen erosion model and the 200 to 600 km O-dominance, density and solar-cycle variation including the Cycle 25 progression chart).

[3] "SPCE 5065 Lesson 3: Human Factors and Bioastronautics," course lecture notes and slides (Canvas), University of Colorado Colorado Springs, 2026 (microgravity deconditioning and exercise countermeasures, isolated-confined-environment group dynamics, astronaut selection and team training).